

Review Article

Rheumatoid Arthritis: Analyzing the Hormonal, Genetic, and Physiological Differences in Male and Females in the Development and Treatment of RA

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Abstract

Rheumatoid Arthritis (RA), an autoimmune disorder, disproportionately impacts women with a 3:1 female to male ratio [1,2]. By analyzing the factors that make up the female (X-linked) genome, hormones, and physiology we can better understand the biological variability that creates this unbalanced ratio and what can be done to further the development of a permanent treatment for RA. This develops the main objective of our article: how do genetic, hormonal, and physiological differences in females contribute to the development of RA? How can this help develop patient-specific treatment and medication?

Introduction

Rheumatoid Arthritis (RA) is a chronic autoimmune disorder that affects synovial joints throughout the body. With RA, your immune system mistakenly targets the healthy tissue lining your joints. This causes an overproduction in the joint cavity's synovial fluid, creating a swollen, inflamed synovial membrane. Over time, this can lead to bone erosion and cartilage degradation because chronic inflammation stimulates bone cells called osteoclasts which break down bone tissue. This process is also called bone resorption. Chronic inflammation can also develop into decreased mobility, bumps over small joints and deformities in the joints of the finger, and the eventual fusing of joints. Effects unrelated to the musculoskeletal system may be fatigue, fever, and impacts on the function of the lungs, heart, skin, nerves, muscles, blood vessels, and kidneys, as the body is forced to compensate [3].

Currently, there is no conclusion on the exact pathogenesis, or the cause/development of RA. Similar to most autoimmune disorders, RA impacts more women than men, and quite disproportionately as well. Additionally, women are more likely to experience RA “worse” than men, or in other words, their symptoms were reported as far more severe [15]. In combination with that, men tended to respond better to temporary, existing treatments although the etiopathogenesis, or the cause/development of RA has not been determined yet. This prompted our research on the biological differences between men and women that affect the diagnosis, development, and treatment of Rheumatoid Arthritis.

Impact of Female Hormones

The higher prevalence of Rheumatoid Arthritis in women suggests that female hormonal factors play a role in the development of the disease. When analyzing the impact of female hormones, we are

specifically addressing fluctuations in serum sex levels that can affect RA [4]. There are two main hormones to analyze: estrogen and progesterone. Estrogen is a group of steroid hormones, including estrone (E1), estradiol (E2), and estriol (E3) that play a vital role in the growth and reproductive development of women. Progesterone is a sex hormone involved with menstruation and prepares the body for pregnancy by thickening the lining of the endometrium [5].

Menstrual Cycle

Before and during menstruation, women have reported a cyclical increase in RA symptoms. While this may be attributed to altered pain perception and “premenstrual tension syndrome,” it is important to factor in the decrease in estrogen levels during this period. On the other hand, post-menstruation, during the ovulatory phase, estrogen levels start to rise. Women report noticeable improvements in symptoms during this phase, implying a direct correlation between estrogen and symptom relief. [2,4]

Pregnancy & Postpartum

During pregnancy, many women experience remission or low disease activity by the third trimester. During pregnancy, high levels of estrogen hormones estriol (E3), produced during the pregnancy, and estetrol (E4), synthesized exclusively by the fetal liver can possibly be cited as reasons for the reduced symptoms. This aligns with the effects of postpartum, in which there is an increased risk of RA worsening or new onset of disease especially after the first trimester postpartum as estrogen levels fall. The effect of pregnancy on RA strongly solidifies the idea that estrogen, or specifically in this scenario E4, could play a role in immune regulation. [2,4,6,7]

Menopause

During menopause, there seems to be a stark increase in the occurrence of RA or the severity of symptoms due to the drop in estrogen. There has also been an inverse correlation in the severity of RA symptoms and the perimenopausal age, as you are less likely to develop RA the later menopause starts. [2,4,6,7]

Estrogen as Treatment

Based on this we can determine that there is a strong relationship between estrogen and the alleviation of RA. Estrogen is noted to have an anti-inflammatory effect in the regulation of helper T cells, and the estrogen prompts the production of anti-inflammatory cytokines such as IL-4, IL-10, and TGF- β [2,6,8]. Thus, this begs the question of how estrogen can be used for treatment. During an experiment, a type II collagen-induced arthritis model (CIA), a model in which type II collagen is injected into a rat or other animal to induce acute joint inflammation, was used to analyze the effect of estrogen. They found that estrogen was able to delay the production of proinflammatory cytokines (TNF- α , (IL)-1 β , and IL-6) and promote the production of anti-inflammatory cytokines (IL-4, IL-10, and TGF- β (Th2 phenotype)), with E3 being the most effective of the estrogens. [9]

Impact of Genetics

The Female Genome

Another factor that may impact the pervasiveness of Rheumatoid Arthritis in women compared to men is the female genome as it differs from that of the male one. Females contain two 'X' chromosomes in their pair of sex chromosomes whereas males contain solely one 'X' chromosome and one 'Y' chromosome. The 'X' chromosome is vastly different from the 'Y' chromosome. For one, it contains 900 active protein coding genes whereas the 'Y' chromosome only contains around 100. Sex chromosomes are not only found in reproductive tissue, but rather in every cell throughout the body. Since chromosomes carry DNA, which hold all of the genetic instructions for cell function, development, growth, and reproduction, the genome of each sex could greatly impact how Rheumatoid Arthritis is caused and how the body reacts to it. [13]

'X' Chromosome Inactivation

Due to the 'X' chromosome having 9 times as many active genes as the 'Y' chromosome, having two operational 'X' chromosomes risks the overproduction of vital proteins in women. To combat this, a molecule of RNA called Xist is produced by the second 'X' chromosome in order to silence the active genes in one of the 'X' chromosomes, thus limiting the production of proteins. This process, called 'X' chromosome inactivation is helpful in ensuring cell survival, however the Xist RNA complex comes with its own complications. Only present in women, Xist binds to the extra X chromosome, causing proteins to bind to it. This leads to the formation of many complexes which trigger a strong immune response [10,14]. The creation of these complexes can cause the female body to create autoantibodies. These findings suggest that a female's genetic makeup causes susceptibility to autoimmunity, and in turn Rheumatoid Arthritis.

Impact of Other Physiological Factors

Musculoskeletal Structure

Rheumatoid Arthritis may appear as more severe in women as far as symptoms and pain may go because of the gap in inherent muscle strength between male and females. This is due to muscle fiber characteristics; a study by the European Journal of Applied Physiology and Occupational Physiology indicates that males tend to have larger muscle fibers than females which is what makes up this gap in strength. For example, Type 1 muscle fibers are 19% larger, Type 2A muscle fibers are 59% larger, and Type 2X muscle fibers are 66% larger in males than females [16]. In Rheumatoid Arthritis, a reliance on greater muscle function in men may cause symptoms caused by inflammation of synovial joints to feel less severe. This can lead to fewer men being diagnosed with RA compared to women, which may be why the female to male ratio contrasts.

Perception of Pain

Females are more likely to experience severe symptoms of RA due to their higher perception of pain. Although a permanent measurement for pain does not exist, a Stanford study [12] found that women reported higher pain scores than men on a scale of 1-10. Although this may not directly correlate with more severe symptoms, it reflects a difference in how women and men experience pain, insinuating varying degrees of pain between men and women in RA.

Nociception

Nociception, another factor involving pain, differs between men and women as well. Nociception is the processing of external stimuli to activate specialized nerve endings called pain receptors. Overall, we found that the stimulus threshold, the minimum intensity of a stimulus to cause a reaction in an organism, is lower in females than males. In men, it was around 18 volts whereas it was around 15 volts for women. This shows that pain tolerance is greater and nociception is lower in males compared to females, which could highlight a difference in how RA is diagnosed. [[11,17](#)]

Conclusion

The main goal of this paper was to better understand how genetic, hormonal, and physiological differences in females contributed to the development and treatment of RA. Thus we conclude that:

1. Constant fluctuations in female hormone levels play a significant role in the onset/remission of Rheumatoid Arthritis symptoms, with estrogen being a key factor in these changes with its anti-inflammatory effects. This also suggests a possibility of estrogen being used to treat RA through hormone therapy.
2. X-chromosome inactivation in women increased chances of them developing an autoimmune disorder.
3. Difference in the musculoskeletal system and nervous system caused women to experience greater severity of symptoms. This opens a path for researchers to look at the development of muscle mass and the cause of differing nociception as mechanisms to mitigate pain in patients.

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